Data Storytelling

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Week 04

Announcement

- Homework 1 is assigned today! Due in two weeks: March 2nd at 11:59 PM
  - Will be uploaded to Canvas and GitHub today
  - I figured that you could all use a little extra time to work on it
- Updated the syllabus to include Llaudet and Imai, Chapter 4 on March 9th
  - Covers prediction, which I think is an important topic to cover in this course

Estimation of Causal Effects with Randomized Experiments

Identifying Causal Relationships

- We want to know the causal effect of X on Y

$$X \to Y$$

- *Examples:*
  - Effect of democracy (X) on economic growth (Y)
  - Effect of class size on student achievement

Inability to Observe Counterfactuals

- True individual causal effect for unit i:

$$\tau_i = Y_i(1) - Y_i(0)$$

- We can only observe one of the potential outcomes for each unit, either the treated outcome (Y(1)) or the control outcome (Y(0)), but not both.
  - Outcome we don't observe is called a `counterfactual`
  - This is known as the `fundamental problem of causal inference`
- Example:
  - We can not observe what the economic growth of the USA would have been if it had not been a democracy, and we can not observe what the economic growth of China would have been if it had been a democracy.

A Confounding Problem

Why not just compare a unit that received the treatment (e.g., a democracy, such as the UK) to a unit that did not receive the treatment (e.g., an autocracy, such as Russia)?

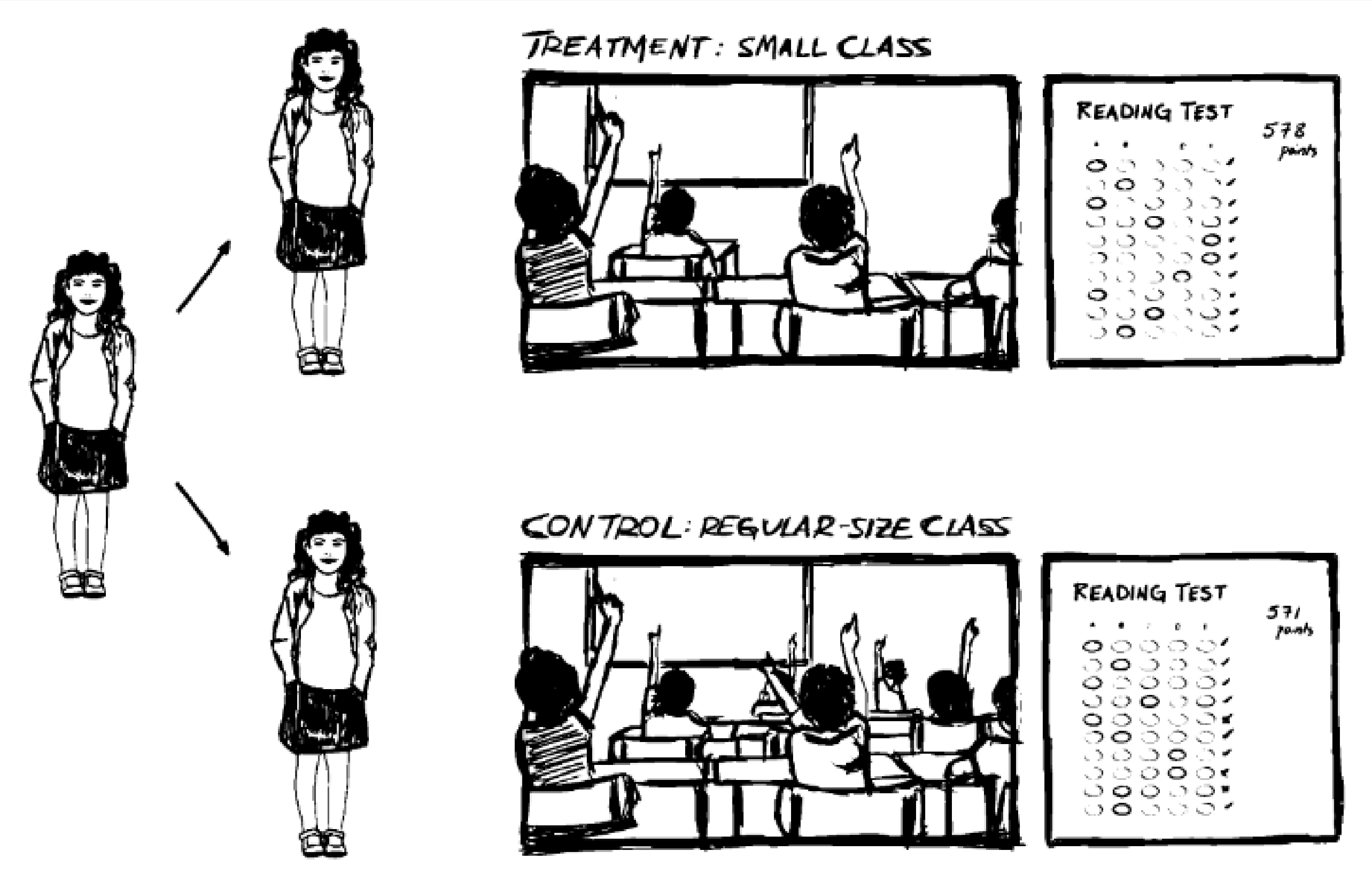
- Existence of `confounding variables` that affect both the treatment and the outcome
  - For example, a country's possession of oil resources may affect both its likelihood of being a democracy and its economic growth
- Possibility of `selection bias` if units self-select into treatment or control groups
  - For example, students who choose to attend a small class may be more motivated than those who choose to attend a large class, which could affect their achievement regardless of class size
- `Reverse causality`, where the outcome affects the treatment rather than the other way around
  - For example, a country's economic growth may affect its likelihood of becoming a democracy rather than democracy affecting economic growth

Example: STAR Experiment

- Let's say we want to know the effect of class size on student achievement.
  - What relationship do we expect to see between class size and student achievement?
- In project STAR (Student/Teacher Achievement Ratio), the following relationship was predicted:

$$\textit{small class} \rightarrow \textit{student performance}$$

Example: STAR Experiment



Ideal Solution to the Problem of Causal Inference

- While we can never observe the true `individual causal effect` it is possible to estimate the `average causal effect` of a treatment
  - Let's say we have units that we can place into a treatment group (T) and a control group (C)
  - If we `randomly assign` units to these groups, then we can assume that any differences in outcomes between the two groups are due to the treatment and not some other confounding variable

$$\text{Average Causal Effect} = E[Y(1)] - E[Y(0)]$$

- This is the basis of *randomized controlled trials* (RCTs), which are considered the gold standard for causal inference

Why Random Assignment Works

- Let's take a group of 100 students
- For each student, we flip a coin to determine whether they are assigned to the treatment group (small class) or the control group (large class)
  - Would we expect any systematic differences in age? Height? Personality? Gender? Etc. between the two groups?
- The random assignment should ensure that the two groups are similar in all respects except for the treatment

Calculating the Treatment Effect

- In the STAR experiment, we can calculate the average treatment effect (ATE) as follows:

$$Average\ effect\ of\ being\ in\ a\ small\ class = E[score(small)] - E[score(big)]$$

- Given that randomization was done correctly, we can be confident that any difference in average scores between the small class and big class groups is due to the effect of class size on student achievement, rather than some other confounding variable.